

Smart Campus Management System

Project Report

Submitted for VITyarthi – Build Your Own Project

1. Introduction

The Smart Campus Management System is a menu-driven Java application for student registration, course management, enrollment, attendance, and campus reporting.

2. Problem Statement

Manual or disconnected records make it difficult to maintain accurate student lists, course rosters, and attendance summaries. This application provides a centralized console workflow.

3. Functional Requirements

Student management; course management; enrollment; attendance marking; student details; course rosters; and campus reports.

4. Non-functional Requirements

Usability through a menu-driven interface; reliability through duplicate and missing-record checks; maintainability through modular classes; and resource efficiency through in-memory collections.

5. System Architecture

The system contains a console interface, application controller, domain records, and in-memory collections. The architecture diagram is included below.

System Architecture

User

Console Menu

Student / Course Management

Enrollment / Attendance

Reporting

6. Design Diagrams

The following pages include system architecture, workflow, use case, sequence, class, and ER diagrams.

Process Workflow

Start → Seed Data → Display Menu

Select Operation → Validate Input

Store / Process Data

Display Result → Return to Menu

Exit

Use Case Diagram

Administrator: Students, Courses, Enrollment, Reports

Faculty: Attendance, Reports

Sequence Diagram

User → Application: Select Enroll

Application → Store: Validate

Store → Application: Result

Application → User: Confirmation

Class Diagram

Main

SmartCampusManagementSystem

Student / Course

Services: Enrollment, Attendance, Reporting

ER Diagram

STUDENT 1 —< ENROLLMENT >— 1 COURSE

COURSE 1 —< ATTENDANCE >— STUDENT

7. Design Decisions & Rationale

Java Records represent compact immutable domain models. Maps provide keyed lookup, sets prevent duplicate enrollment, TreeMap organizes attendance by date, and LocalDate represents attendance dates.

8. Implementation Details

The project uses Java 17+, Scanner, switch expressions, collections, streams, exception handling, and a package-based source structure.

9. Screenshots / Results

The screenshots show the main menu, campus report, and student details output.

```
Smart Campus Management System – Console Output
```

```
=== SMART CAMPUS MANAGEMENT SYSTEM ===
```

1. Add student
2. Add course
3. Enroll student
4. Mark attendance
5. View student details
6. View course roster
7. Campus report
8. Exit

```
Smart Campus Management System – Console Output
```

```
=== CAMPUS REPORT ===
```

```
Students: 2 | Courses: 2 | Enrollments: 2  
CS101: 2 enrolled, attendance 100.0% (1 sessions)  
MA201: 0 enrolled, attendance 0.0% (0 sessions)
```

```
Smart Campus Management System – Console Output
```

```
Student ID: S001  
Aisha Khan | aisha@campus.edu | Computer Science  
Enrolled courses:  
- CS101: Programming Fundamentals
```

10. Testing Approach

The test plan covers valid operations, duplicate IDs, invalid capacities, missing records, capacity limits, attendance, and reporting.

11. Challenges Faced

The main challenge is maintaining consistent relationships between students, courses, enrollments, and attendance while validating user input.

12. Learnings & Key Takeaways

The project demonstrates object-oriented programming, records, collections, maps and sets, exception handling, date handling, modularization, and reporting.

13. Future Enhancements

Database integration, role-based login, GUI/web interface, attendance history, CSV export, search filters, and automated unit tests.

14. References

VITyarthi Build Your Own Project submission guidelines and Java platform documentation.